Preliminary Report
Hurricane Edouard
19 August - 3 September 1996

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Edouard, the strongest tropical cyclone of the 1996 Atlantic season, was a prototypical Cape Verde hurricane. It had a very long track, and maintained category three or greater intensity on the Saffir/Simpson Hurricane Scale for nearly eight days. Edouard brushed southeastern New England as it recurved out to sea.

a. Synoptic History

Edouard originated from a tropical wave that was already well-marked by a spiral-shaped mass of convective clouds while moving across western Africa on 17-18 August. The wave crossed the west coast of Africa early on the 19th, accompanied by a 45-knot mid-tropospheric jet seen in rawinsonde data. Observations from Dakar and nearby stations showed thunderstorms and squalls, along with 24-hour surface pressure falls on the order of 3 to 4 mb as the wave passed. Soon after entering the eastern tropical Atlantic, ship reports showed the presence of a large surface circulation. It is estimated that a tropical depression formed around 1800 UTC on 19 August, centered about 300 n mi southeast of the Cape Verde Islands, as shown in the post-analysis "best track" in Table 1 and Fig. 1. This was the first in a series of four tropical cyclones that would form over the eastern Atlantic from waves that moved off the west coast of Africa during a two-week span in late August and early September of 1996. Three of these systems (Edouard, Fran, and Hortense) eventually became category three (or stronger) hurricanes.

Initially, it appeared that the westward-moving tropical cyclone would soon take a northwestward turn in response to a weakness in the subtropical ridge over the eastern Atlantic. However, the subtropical ridge remained strong enough to the north of the system to keep it on a generally westward track into the central tropical Atlantic. Higher-level winds favored intensification of the cyclone, as an upper-tropospheric anticyclone became well established over the area. The system became Tropical Storm Edouard early on 22 August, and strengthened into a hurricane around

1200 UTC the following day, when a banding-type eye was noted in satellite pictures.

When the hurricane neared 45°W longitude on the 24th, a deep-layer cyclone to the east of Bermuda began to create a weakness in the subtropical ridge. In response to this, Edouard's direction of motion changed from westward to west-northwestward. Meanwhile, intensification continued, and Edouard's winds strengthened to 100 knots on the 24th and to 125 knots on the 25th, making it a category four hurricane. The latter wind speed was the maximum intensity, and a similar wind speed is estimated on the 26th and also around 0000 UTC on the 28th. From the 26th to the 28th, some fluctuations in intensity were noted, apparently as the result of eyewall replacement cycles and occasional doses of stronger vertical shear over the area. Nonetheless, Edouard maintained 115 knot or greater winds throughout the above period . The final deepening episode in Edouard was observed late on 29-30 August. During that event, *three* concentric eyewalls were indicated by aerial reconnaissance observations. Overall, Edouard remained a powerful, 100-knot or stronger hurricane for a very long time - from 24 August until early on 1 September.

Edouard moved relentlessly towards the west-northwest, at around 12 knots, until the 29th of August. This track kept the hurricane well to the northeast and north of the islands of the northeastern Caribbean Sea. On the 29th, a mid-tropospheric trough became established near the U.S. east coast, creating a more northward steering component for Edouard. Slowing its forward speed slightly, the hurricane turned northwestward, and then northward, while gradually weakening. The cyclone passed about midway between Cape Hatteras and Bermuda on 1 September, and then started moving slightly east of north. Late on the 1st, the hurricane wobbled toward the north, in the general direction of southeastern New England. However, early on the 2nd, Edouard veered sharply toward the northeast, and the center of the hurricane passed about 75 n mi southeast of Nantucket island around 0900 UTC, the closest point of approach to the United States. Maximum winds had diminished to near 70 knots by that time.

Edouard weakened to a tropical storm near 0000 UTC on the 3rd, and became extratropical shortly thereafter. The storm's motion became east-northeastward, keeping the center south of Nova Scotia, and, later, well offshore of Newfoundland. Edouard's remnant low was drawn around and into the circulation of a larger extratropical cyclone on the 6th, and was absorbed by this bigger system by 0000

UTC 7 September.

b. Meteorological Statistics

Figures 2 and 3 depict the curves of minimum central sea-level pressure and maximum one-minute average "surface" (10 meters above ground level) wind speed, respectively, as a function of time. Also plotted are the observations on which the curves are based, consisting of aircraft reconnaissance data, Dvorak-technique estimates using satellite imagery, and fixes from synoptic analyses.

Most of the aircraft reconnaissance flights into Edouard were accomplished by the "Hurricane Hunters" of the U.S. Air Force Reserves. The Hurricane Hunters flew 15 missions, and made 66 center fixes. NOAA aircraft provided four additional fixes. The highest wind speed reported was 140 knots (at 700 mb) at 0003 UTC 28 August. Lowest central pressure reported was 934 mb at 1727 UTC 30 August. However, the highest wind reported by aircraft around that time was 134 knots. Subjective and objective Dvorak intensity estimates indicate that Edouard was stronger on 25-26 August, and also at 0000 UTC 28 August, than it was at the time of the minimum aircraft-reported pressure. At the latter time, the hurricane appeared considerably less well-organized on satellite images than on the earlier days.

Since Edouard crossed over the New York shipping channels, there was a large number of encounters by vessels at sea with this hurricane. Table 2 lists ship reports of tropical storm force or greater wind speeds associated with Edouard.

The hurricane came close enough to New England to produce sustained winds of tropical storm force at Nantucket Island and the Cape Cod area. Wind gusts to hurricane force were reported at Nantucket. Table 3 lists selected surface observations from Massachusetts, New Hampshire and Maine. In addition, there were unofficial reports of wind gusts to 90 mph at Nantucket, 80 mph at Martha's Vineyard, and 77 mph on Cape Cod.

Large swells, minor beach erosion, and some coastal flooding, presumably minor as well, occurred along the coast from North Carolina northward through Maine.

c. Casualty and Damage Statistics

Two deaths have been directly attributed to Edouard. A 71-year old man died

when his boat capsized in heavy surf in Great Egg Harbor Inlet, south of Atlantic City, New Jersey. A 28-year old man drowned while surfing at Lavallette, northeast of Tom's River, New Jersey. Additionally, a 44-year old man suffered a broken neck (but survived) while surfing near Atlantic City. Overall, the effects of Edouard on land were apparently minor. Most of the damage was to boats at Martha's Vineyard and Nantucket.

d. Forecast and Warning Critique

In general, the track forecasts for Edouard were excellent. This may be attributable to the objective guidance, which was quite accurate in most cases. Table 4 lists the average track model and official forecast errors for Edouard. One can see that the average official forecast errors are spectacularly low, in comparison to the most recent ten-year averages. The same can be said for the most oft-used objective track forecast techniques, viz the GFDI, BAMD, and A90E (NHC90). The U.K. Met. Office model, UKMI, also had rather low average track forecast errors.

NHC advisories had large underforecasts of intensity, up to 65 knots at 72 hours, during the early stages of Edouard. Intensity was significantly overforecast, on the order of 25 to 45 knots in 48 to 72 hours, on 30-31 August. Otherwise, the intensity predictions were good.

Since the official forecast tracks were bringing Edouard close to the U.S. east coast, and there was the usual uncertainty in these forecasts, watches and warnings were required from the mid-Atlantic states northward. Table 5 summarizes these watches and warnings. A hurricane warning was posted for portions of Rhode Island and Massachusetts at 0900 UTC 1 September. Although sustained hurricane force winds did not occur in these areas, sustained tropical storm force winds, with gusts to hurricane force, were observed in the eastern end of the hurricane warning area 18 to 24 hours after the issuance of the warning. At 0900 UTC on the 2nd, the hurricane warning was changed to a tropical storm warning over southeastern New England, even though it was clear that Edouard was bypassing the area. The reason for downgrading the warning (as opposed to lowering all warnings) was that Edouard was slow to exit, and strong winds were likely to continue lashing the coast in the vicinity of Cape Cod during the day. All U.S. warnings were dropped at 0000 UTC 3 September, by which time Edouard was nearing Nova Scotia.

Table 1. Best track, Hurricane Edouard, 19 August - 3 September, 1996

| Date/Time (UTC) | Position | | Pressure (mb) | Wind Speed (kt) | Stage |
|---|---|---|---|---|---|
| | Lat. (°N) | Lon. (°W) | | | |
| 19/1800 | 12.4 | 19.9 | 1007 | 25 | tropical depression |
| 20/0000 | 12.4 | 20.7 | 1007 | 25 | " |
| 0600 | 12.4 | 21.6 | 1007 | 25 | " |
| 1200 | 12.5 | 22.6 | 1007 | 25 | " |
| 1800 | 12.6 | 23.8 | 1007 | 25 | " |
| 21/0000 | 12.6 | 25.3 | 1007 | 25 | " |
| 0600 | 12.8 | 26.7 | 1007 | 25 | " |
| 1200 | 12.9 | 27.9 | 1007 | 25 | " |
| 1800 | 13.0 | 29.2 | 1006 | 25 | " |
| 22/0000 | 13.1 | 30.3 | 1006 | 30 | " |
| 0600 | 13.2 | 31.6 | 1005 | 35 | tropical storm |
| 1200 | 13.4 | 32.9 | 1003 | 40 | " |
| 1800 | 13.5 | 34.4 | 1002 | 40 | " |
| 23/0000 | 13.7 | 35.8 | 1000 | 45 | " |
| 0600 | 13.8 | 37.0 | 996 | 50 | " |
| 1200 | 13.9 | 38.3 | 988 | 65 | hurricane |
| 1800 | 14.0 | 39.8 | 986 | 65 | " |
| 24/0000 | 14.1 | 41.1 | 983 | 70 | " |
| 0600 | 14.1 | 42.4 | 980 | 75 | " |
| 1200 | 14.4 | 43.6 | 970 | 90 | " |
| 1800 | 14.6 | 44.7 | 960 | 100 | " |
| 25/0000 | 15.0 | 45.8 | 942 | 120 | " |
| 0600 | 15.4 | 47.0 | 933 | 125 | " |
| 1200 | 15.9 | 48.3 | 934 | 125 | " |
| 1800 | 16.5 | 49.6 | 935 | 125 | " |
| 26/0000 | 17.0 | 50.9 | 936 | 125 | " |
| 0600 | 17.5 | 52.0 | 936 | 125 | " |
| 1200 | 17.9 | 53.2 | 937 | 125 | " |
| 1800 | 18.4 | 54.5 | 938 | 125 | " |
| 27/0000 | 18.9 | 55.6 | 942 | 125 | " |
| 0600 | 19.5 | 56.9 | 948 | 120 | " |
| 1200 | 20.0 | 58.1 | 952 | 115 | " |
| 1800 | 20.4 | 59.3 | 951 | 115 | " |
| 28/0000 | 20.9 | 60.4 | 944 | 125 | " |
| 0600 | 21.3 | 61.7 | 952 | 115 | " |
| 1200 | 21.9 | 63.2 | 960 | 110 | " |
| 1800 | 22.3 | 64.6 | 956 | 110 | " |

Table 1 (continued). Best track, Hurricane Edouard, 19 August - 3 September, 1996

| | | | | | |
|---|---|---|---|---|---|
| 29/0000 | 22.9 | 65.9 | 957 | 110 | " |
| 0600 | 23.4 | 67.1 | 961 | 110 | " |
| 1200 | 24.0 | 68.1 | 950 | 115 | " |
| 1800 | 24.7 | 69.0 | 948 | 120 | " |
| 30/0000 | 25.5 | 69.5 | 941 | 120 | " |
| 0600 | 26.4 | 69.9 | 939 | 120 | " |
| 1200 | 27.4 | 70.2 | 938 | 120 | " |
| 1800 | 28.5 | 70.5 | 934 | 120 | " |
| 31/0000 | 29.5 | 70.5 | 944 | 110 | " |
| 0600 | 30.4 | 70.6 | 950 | 105 | " |
| 1200 | 31.6 | 70.3 | 952 | 105 | " |
| 1800 | 32.7 | 70.1 | 953 | 100 | " |
| 01/0000 | 34.0 | 70.1 | 959 | 100 | " |
| 0600 | 35.2 | 70.1 | 958 | 100 | " |
| 1200 | 36.5 | 70.2 | 958 | 95 | " |
| 1800 | 37.5 | 70.0 | 960 | 85 | " |
| 02/0000 | 38.7 | 69.5 | 964 | 80 | " |
| 0600 | 39.8 | 69.4 | 961 | 70 | " |
| 1200 | 40.5 | 68.3 | 962 | 70 | " |
| 1800 | 41.3 | 67.3 | 972 | 65 | " |
| 03/0000 | 42.0 | 66.8 | 978 | 60 | tropical storm |
| 0600 | 42.2 | 66.0 | 985 | 55 | extratropical |
| 1200 | 42.5 | 65.0 | 992 | 50 | " |
| 1800 | 42.7 | 63.0 | 995 | 50 | " |
| 04/0000 | 43.2 | 59.8 | 997 | 50 | " |
| 0600 | 43.3 | 57.5 | 994 | 50 | " |
| 1200 | 43.4 | 55.5 | 994 | 50 | " |
| 1800 | 43.5 | 53.5 | 995 | 50 | " |
| 05/0000 | 43.5 | 51.5 | 995 | 50 | " |
| 0600 | 43.5 | 50.0 | 994 | 45 | " |
| 1200 | 43.5 | 48.5 | 994 | 45 | " |
| 1800 | 43.7 | 47.5 | 995 | 45 | " |
| 06/0000 | 44.5 | 46.0 | 995 | 40 | " |
| 0600 | 46.0 | 44.0 | 994 | 40 | " |
| 1200 | 48.0 | 43.0 | 992 | 40 | " |
| 1800 | 50.0 | 41.0 | 990 | 40 | " |
| 07/0000 | | | | | absorbed by larger low |

| | | | | | |
|---|---|---|---|---|---|
| 25/0600 | 15.4 | 47.0 | 933 | 125 | minimum pressure |

Table 2. Ship reports of 34 knots or higher wind speed associated with Hurricane Edouard, August/September 1996.

| date/time (UTC) | ship call sign | latitude (°N) | longitude (°W) | wind dir/ speed(knots) | pressure (mb) |
|---|---|---|---|---|---|
| 26/0600 | DHEE | 11.5 | 53.5 | 210/44 | 1012.0 |
| 30/1800 | OZBL2 | 29.8 | 69.1 | 090/64 | 1010.0 |
| 31/0000 | WZJF | 33.2 | 73.7 | 050/35 | 1011.0 |
| 31/0600 | OZBL2 | 32.1 | 71.1 | 090/60 | 1010.0 |
| 31/1200 | 9VGL | 29.8 | 67.3 | 150/36 | 1009.0 |
| 31/1800 | DDQS | 29.2 | 67.7 | 190/37 | 1010.9 |
| 01/0000 | SHIP | 27.8 | 70.1 | 270/41 | 1010.3 |
| 01/0000 | 9VGL | 32.2 | 66.6 | 150/34 | 1009.0 |
| 01/0600 | WXKM | 33.1 | 66.0 | 150/35 | 1012.0 |
| 01/0600 | ZCBC3 | 35.0 | 68.0 | 180/37 | 1001.4 |
| 01/0600 | 9VGL | 33.4 | 66.5 | 150/36 | 1009.0 |
| 01/0900 | PGAF | 36.0 | 65.3 | 120/35 | 1010.0 |
| 01/1200 | KMJL | 38.7 | 69.5 | 060/58 | 1005.9 |
| 01/1200 | PJJU | 36.8 | 66.0 | 130/40 | 1007.0 |
| 01/1200 | ZCBC3 | 36.3 | 67.3 | 140/37 | 1001.6 |
| 01/1200 | 9VGL | 34.6 | 66.5 | 150/38 | 1007.8 |
| 01/1500 | KMJL | 38.7 | 69.1 | 060/68 | 1000.8 |
| 01/1500 | PEBP | 36.4 | 73.4 | 010/41 | 1007.2 |
| 01/1500 | PGAF | 35.2 | 65.3 | 150/35 | 1010.8 |
| 01/1800 | KMJL | 38.7 | 68.9 | 110/40 | 999.1 |
| 01/1800 | LXWB | 35.9 | 64.5 | 160/34 | 1014.5 |
| 01/1800 | PEBP | 36.1 | 73.6 | 360/39 | 1009.0 |
| 01/1800 | PGAF | 34.7 | 65.3 | 140/35 | 1012.2 |
| 01/1800 | SHIP | 35.4 | 74.8 | 360/58 | - |
| 01/1800 | V2MF | 39.5 | 66.6 | 120/40 | 1010.0 |
| 01/1800 | ZCBC3 | 37.0 | 68.0 | 180/44 | 1006.4 |
| 01/1800 | 9VGL | 35.3 | 67.3 | 190/40 | 1006.0 |
| 01/2100 | DV2MFB | 39.4 | 66.5 | 140/45 | 1006.2 |
| 01/2100 | KMJL | 38.7 | 68.4 | 130/60 | 995.8 |
| 01/2100 | ZCBC3 | 36.5 | 68.0 | 180/37 | 996.5 |
| 02/0000 | C6MZ8 | 40.4 | 70.3 | 050/40 | 1003.2 |

Table 2 (cont.). Ship reports of 34 knots or higher wind speed associated with Hurricane Edouard, August/ September 1996.

| date/time (UTC) | ship call sign | latitude (°N) | longitude (°W) | wind dir/ speed(knots) | pressure (mb) |
|---|---|---|---|---|---|
| 02/0000 | KMJL | 38.8 | 68.0 | 140/70 | 994.0 |
| 02/0000 | LXWB | 36.2 | 65.6 | 160/40 | 1011.2 |
| 02/0000 | SKOZ | 40.5 | 71.1 | 040/37 | 1003.2 |
| 02/0000 | ZCBC3 | 36.5 | 68.1 | 180/37 | 997.4 |
| 02/0000 | 9VGL | 36.0 | 68.1 | 210/42 | 994.2 |
| 02/0300 | KMJL | 38.8 | 67.7 | 150/70 | 995.1 |
| 02/0300 | SKOZ | 40.2 | 72.1 | 030/37 | 1003.5 |
| 02/0300 | 4XGR | 39.7 | 66.0 | 130/34 | 1003.0 |
| 02/0600 | KMJL | 38.7 | 67.5 | 190/70 | 997.5 |
| 02/0600 | LXWB | 36.5 | 66.7 | 220/43 | 1009.2 |
| 02/0600 | NFMK | 41.9 | 70.2 | 040/35 | 1006.5 |
| 02/0600 | NJPJ | 41.6 | 70.4 | 040/40 | 1013.5 |
| 02/0600 | SKOZ | 40.0 | 72.4 | 360/37 | 1003.5 |
| 02/0600 | VCRJ | 36.8 | 64.1 | 180/40 | 1013.3 |
| 02/0600 | V2MF | 39.1 | 66.0 | 180/45 | 1001.5 |
| 02/0600 | 4XGR | 39.4 | 66.0 | 160/50 | 1001.5 |
| 02/0600 | 9VGL | 36.5 | 68.7 | 250/36 | 1004.5 |
| 02/0900 | KMJL | 38.5 | 67.2 | 210/45 | 999.9 |
| 02/0900 | V2MF | 39.0 | 66.0 | 210/49 | 1000.0 |
| 02/0900 | 4XGR | 39.0 | 65.9 | 160/45 | 1001.0 |
| 02/1200 | C6MZ8 | 38.7 | 70.5 | 320/55 | 1002.1 |
| 02/1200 | KMJL | 38.3 | 66.9 | 200/45 | 1003.9 |
| 02/1200 | LXWB | 36.8 | 67.3 | 230/39 | 1010.7 |
| 02/1200 | NFMK | 41.9 | 70.3 | 010/42 | 1002.5 |
| 02/1200 | V2MF | 39.0 | 66.1 | 180/48 | 1000.0 |
| 02/1200 | ZCBC3 | 38.0 | 69.4 | 250/37 | 1005.4 |
| 02/1200 | 4XGR | 39.2 | 66.6 | 200/45 | 997.5 |
| 02/1200 | 9VGL | 37.1 | 69.3 | 290/39 | 1008.0 |
| 02/1500 | KMJL | 38.1 | 67.0 | 220/45 | 1005.0 |
| 02/1500 | V2MF | 38.7 | 66.0 | 180/47 | 1001.0 |
| 02/1500 | 4XGR | 39.5 | 67.2 | 220/54 | 993.0 |

Table 2 (cont.). Ship reports of 34 knots or higher wind speed associated with Hurricane Edouard, August/September 1996.

| date/time (UTC) | ship call sign | latitude (°N) | longitude (°W) | wind dir/ speed(knots) | pressure (mb) |
|---|---|---|---|---|---|
| 02/1800 | C6MZ8 | 38.4 | 69.3 | 310/37 | 1005.5 |
| 02/1800 | LXWB | 36.9 | 68.2 | 280/34 | 1011.2 |
| 02/1800 | NJPJ | 42.0 | 70.4 | 010/40 | 1004.1 |
| 02/1800 | VYQJ | 44.6 | 66.4 | 040/35 | 1009.0 |
| 02/1800 | V2MF | 38.5 | 65.8 | 200/47 | 1002.0 |
| 02/1800 | ZCBC3 | 39.0 | 70.4 | 330/37 | 1007.4 |
| 02/1800 | 4XGR | 39.6 | 68.0 | 270/45 | 996.0 |
| 02/1800 | 9VGL | 37.5 | 69.7 | 300/44 | 1010.0 |
| 02/2100 | VYQJ | 44.0 | 66.7 | 030/39 | 1003.5 |
| 02/2100 | V2MF | 38.2 | 65.5 | 230/45 | 1005.0 |
| 03/0000 | C6MZ8 | 38.7 | 67.8 | 290/47 | 1000.6 |
| 03/0000 | ELRZ8 | 36.8 | 65.8 | 240/40 | 1011.0 |
| 03/0000 | KMJL | 38.2 | 67.7 | 280/40 | 1008.0 |
| 03/0000 | VCRJ | 41.2 | 65.6 | 200/46 | 994.3 |
| 03/0000 | VCRZ | 43.1 | 70.3 | 330/35 | 1009.5 |
| 03/0000 | VYQJ | 43.5 | 66.3 | 060/38 | 996.0 |
| 03/0000 | V2MF | 37.7 | 65.1 | 230/41 | 1007.0 |
| 03/0000 | 3ETG9 | 41.3 | 69.4 | 020/35 | 1023.0 |

Table 3. Hurricane Edouard selected surface observations, September, 1996.

| Location | Press. (mb) | Date/ time (UTC) | Sustained wind (kts)[a] | Peak gust (kts) | Date /time (UTC)[b] | Storm surge (ft)[c] | Storm tide (ft)[d] | total rain (in) |
|---|---|---|---|---|---|---|---|---|
| **Massachusetts** | | | | | | | | |
| Brant Point Coast Guard | | | 50 | 69 | 02/1900 | | | |
| Buzzards Bay buoy (44028) | | | 39 | 54 | 02/1500 | | | |
| Chatham upper air site | | | | 62 | 02/1117 | | | |
| Falmouth | 1001.7 | 02/1021 | | | | | | 4.97 |
| Georges Bank buoy (44011) | 986.8 | 02/1700 | 39 | 50 | 02/1700 | | | |
| Hyannis | 1001.0 | 02/1039 | | | | | | 5.20 |
| Martha's Vineyard | 1000.0 | 02/0945 | | 47 | | | | |
| Menemsha Coast Guard | | | 55 | | 02/1500 | | | |
| Nantucket buoy (44008) | 979.9 | 02/0500 | 39 | 50 | 02/0500 | | | |
| Nantucket Island | | | | | | 2.1 | 5.6 | |
| Nantucket tower (ACK) | 995.9 | 02/0945 | 45 | 65 | 02/0945 | | | |
| New Bedford ASOS | | | 35 | 46 | 02/1029 | | | |
| Pocasset | | | | | | | | 4.70 |
| West Dennis | | | | | | | | 6.37 |
| West Yarmouth | | | | | | | | 4.75 |
| **New Hampshire** | | | | | | | | |
| Portsmouth | | | | | | | | 1.00 |
| **Maine** | | | | | | | | |
| Eastport | | | | | | | | 1.23 |
| Eliot | | | | | | | | 1.17 |
| Matinicus Rock C-MAN | | | 35 | 41 | 02/1900 | | | |
| Mt. Desert Rock C-MAN | | | 41 | 47 | 02/2000 | | | |
| Westbrook | | | | | | | | 1.16 |

[a]NWS standard averaging period is 1 min; ASOS and C-MAN are 2 min; buoys are 8 min.
[b]Date/time is for sustained wind when both sustained and gust are listed.
[c]Storm surge is water height above normal astronomical tide level.
[d]Storm tide is water height above NGVD.

Table 4

**Preliminary forecast evaluation of Hurricane Edouard**
**Heterogeneous sample**

(Errors in nautical miles for tropical storm
and hurricane stages with number
of forecasts in parenthesis)

| Technique | Period (hours) | | | | |
|---|---|---|---|---|---|
| | 12 | 24 | 36 | 48 | 72 |
| CLIP | 26 (46) | 50 (44) | 81 (42) | 114 (40) | 168 (36) |
| GFDI | 32 (46) | 51 (44) | 66 (42) | 80 (40) | 147 (36) |
| GFDL* | 30 (23) | 51 (22) | 67 (21) | 78 (20) | 111 (18) |
| VBAR | 23 (41) | 40 (39) | 58 (37) | 84 (35) | 189 (31) |
| LBAR | 25 (46) | 43 (44) | 65 (42) | 87 (40) | 156 (36) |
| AVNI | 43 (44) | 84 (42) | 130 (41) | 176 (39) | 279 (36) |
| BAMD | 27 (36) | 47 (44) | 67 (42) | 88 (40) | 151 (36) |
| BAMM | 35 (46) | 70 (44) | 111 (42) | 155 (40) | 228 (36) |
| BAMS | 53 (46) | 109 (44) | 168 (42) | 219 (40) | 293 (36) |
| A90E | 26 (46) | 46 (44) | 73 (42) | 99 (40) | 180 (36) |
| NGPI | 55 (18) | 123 (16) | 215 (14) | 330 (12) | 552 ( 8) |
| UKMI | 44 (28) | 66 (28) | 71 (26) | 96 (24) | 171 (21) |
| | | | | | |
| NHC OFFICIAL | 25 (46) | 46 (44) | 66 (42) | 84 (40) | 118 (46) |
| NHC OFFICIAL 1986-1995 10-year average | 49 (1670) | 93 (1484) | 136 (1314) | 181 (1155) | 273 (882) |

* GFDL output not available until after forecast issued. VBAR output sometimes not available until after forecast issued.

Table 5. Watch and warning summary, Hurricane Edouard, August/September 1996.

| Date/time (UTC) | Action | Location |
|---|---|---|
| 30/2100 | hurricane watch and tropical storm warning issued | Cape Lookout, North Carolina northward to Cape Henlopen, Delaware including the Pamlico and Albermarle Sounds |
| 31/2100 | hurricane watch extended northward | north of Cape Henlopen, Delaware to Plymouth, Massachusetts |
| 31/2100 | hurricane watch and tropical storm warning discontinued | south of Cape Charles, Virginia |
| 1/0300 | hurricane watch extended northward | north of Plymouth, Massachusetts to Merrimack River, Massachusetts |
| 1/0900 | hurricane warning issued | Watch Hill, Rhode Island to Merrimack River, Massachusetts |
| 1/0900 | hurricane watch extended northward | north of Merrimack River, Massachusetts to Eastport, Maine |
| 1/0900 | tropical storm warning issued | Fire Island Inlet, Long Island, New York to Watch Hill, Rhode Island including Long Island Sound east of Port Jefferson Harbor |
| 1/0900 | all watches and warnings discontinued | south of Brigantine New Jersey |
| 1/1500 | tropical storm warning issued | Merrimack River , Massachusetts to Eastport, Maine |
| 1/2100 | hurricane watch discontinued | New Jersey Coast and Long Island west of Fire Island Inlet |

Table 5 (cont.). Watch and warning summary, Hurricane Edouard, August/September 1996.

| | | |
|---|---|---|
| 2/0300 | hurricane watch discontinued | Long Island, Long Island Sound and the coast of New England west of Watch Hill, Rhode Island |
| 2/0900 | hurricane warning changed to tropical storm warning | Watch Hill, Rhode Island to Woods Hole, Massachusetts and from Plymouth, Massachusetts to Merrimack River, Massachusetts |
| 2/0900 | hurricane watch discontinued | north of Merrimack River, Massachusetts |
| 2/1500 | hurricane warning changed to tropical storm warning | Woods Hole, Massachusetts to Plymouth, Massachusetts |
| 2/1500 | watches and warnings discontinued | west of Woods Hole, Massachusetts and north of Plymouth Massachusetts |
| 3/0000 | tropical storm warning discontinued | Woods Hole, Massachusetts to Plymouth, Massachusetts |

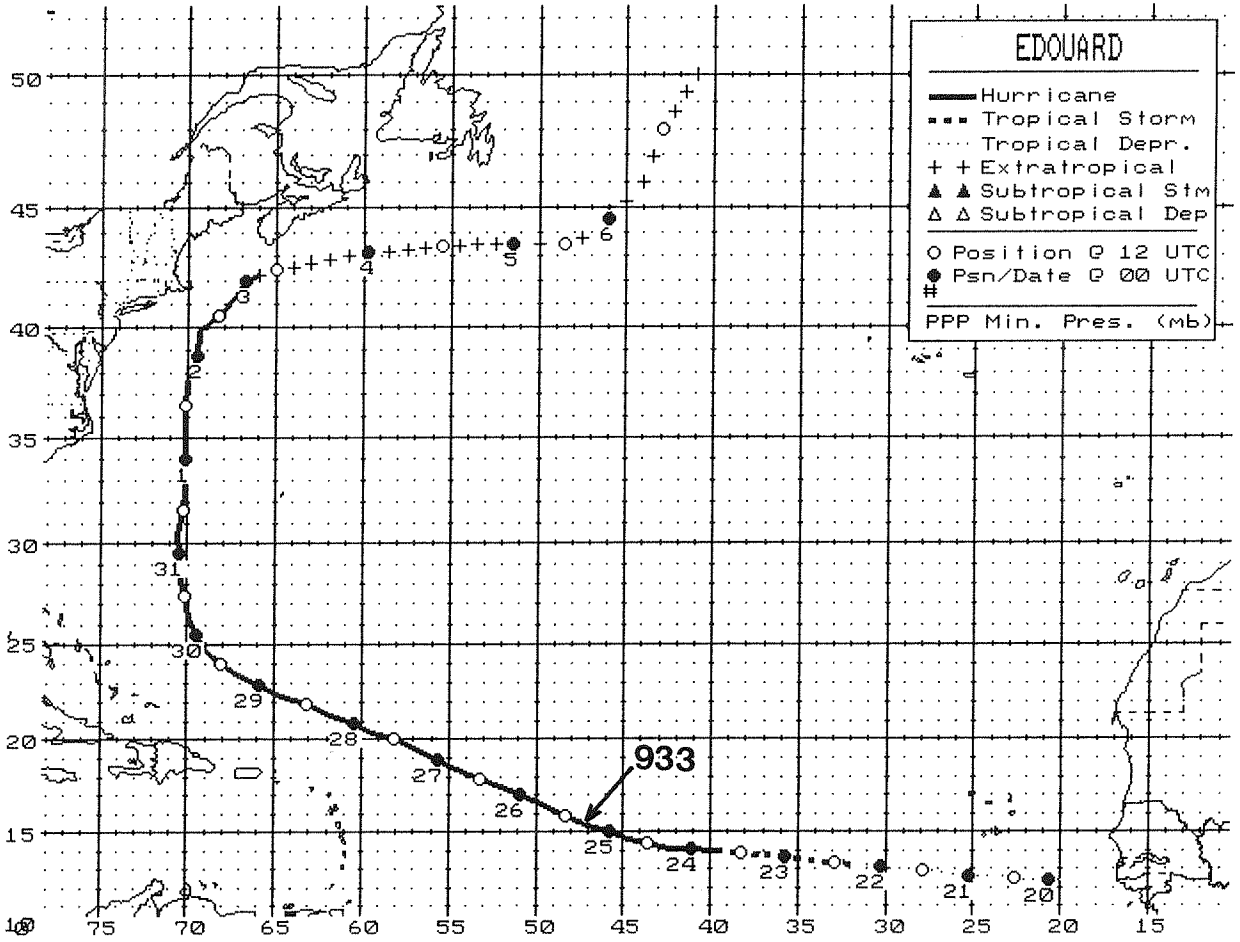


Figure 1. Best track positions for Hurricane Edouard, August/September 1996.

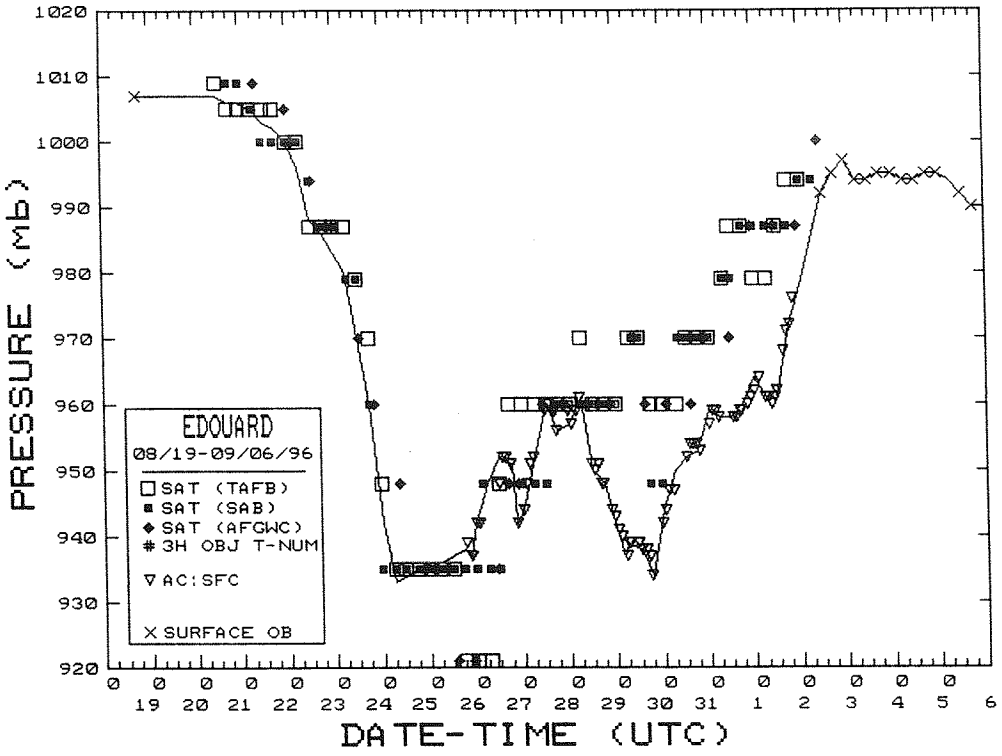


Figure 2. Best track central pressure curve for Hurricane Edouard, August/September

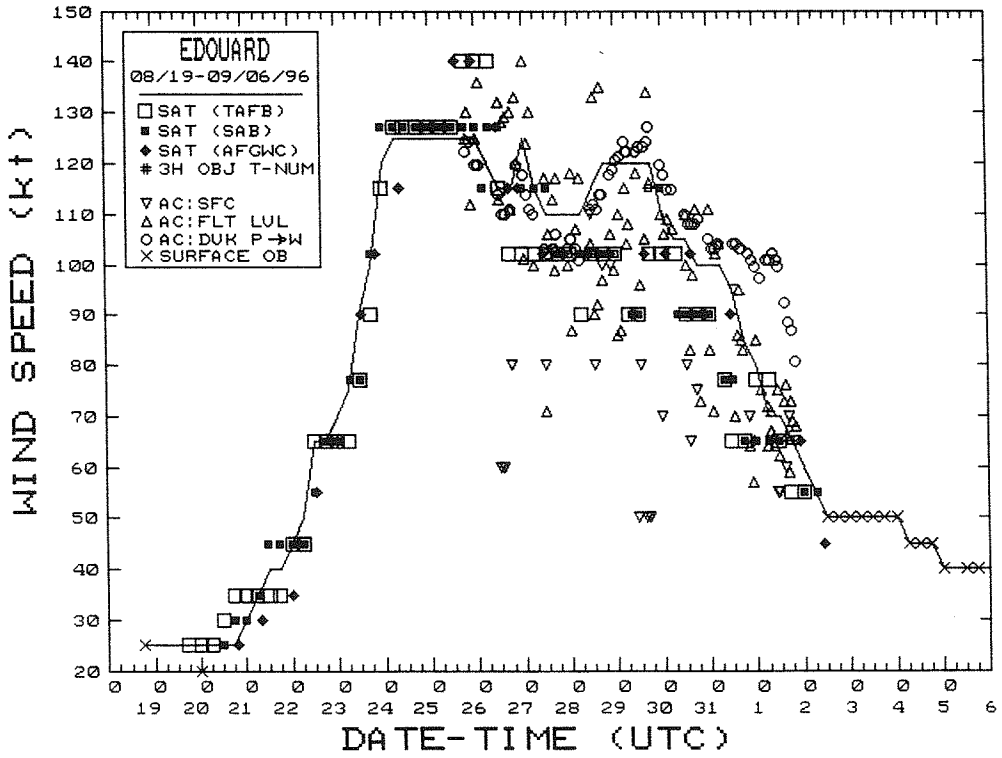


Figure 3. Best track maximum one-minute wind speed curve for Hurricane Edouard, August/September 1996.